

Sagar Prajapati

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LinkedIn | GitHub | GeeksforGeeks | LeetCode

Summary

AI/ML Engineer with hands-on experience in deep learning, computer vision, RAG, LLM applications, and AI systems engineering. Built real-time CNN-based detection systems, optimized PyTorch inference on NVIDIA DGX systems, and developed RAG applications using LangChain, FAISS, and Sentence Transformers. Strong backend and systems background with Go, Node.js, Redis, Docker, and distributed systems, including a production-grade distributed key-value store built from scratch. IEEE conference author for StegIntel (INSECT 2026). Seeking AI/ML Engineer / SDE-1 opportunities.

Education

D.Y. Patil International University, Pune <i>B.Tech in Computer Science (AI & ML) — GPA: 8.9</i>	Aug 2022 – May 2026
SVKM's Mithibai College of Arts & Science, Mumbai <i>Higher Secondary (Science)</i>	Jun 2020 – May 2022
New Model English High School, Mumbai <i>SSC — 91.20%</i>	Jun 2009 – Apr 2020

Achievements

- Secured 4 Pre-Placement Offers (PPOs) during final year of B.Tech.
- Amazon SDE-1 – Advanced to the Online Assessment stage through off-campus recruitment.
- Student of the Year (2019–20) – New Model English High School — Rank 1 with 91.20
- Student of the Year (2023–24) – DY Patil International University — Rank 2 with 9.71 CGPA.
- GeeksforGeeks Rank Holder – 540+ DSA Problems Solved in C++ — Institutional Rank 3.
- IEEE INSECT 2026 Author – Conference Paper: StegIntel.

Experience

SDE Intern <i>CheckRed Security, Pune</i>	Feb 2026 – May 2026
– Automated RPM packaging pipelines for Linux (RHEL, Rocky) using rpmbuild, spec files, and Bash, enabling reproducible and versioned builds. – Developed OVA build workflows for secure, pre-configured VM images, improving deployment scalability. – Debugged build and dependency failures via log analysis, increasing pipeline reliability. – Integrated CI/CD pipelines to automate build, test, and release processes. – Standardized build environments, reducing deployment inconsistencies across releases.	
AI Research Intern <i>COEP Technological University, Pune</i>	May 2025 – Aug 2025
– Trained a custom CNN for chit-paper detection and behavior tracking, improving accuracy by 27%. – Optimized inference speed by 40% using PyTorch tuning on NVIDIA DGX systems. – Automated dataset annotation workflows using Roboflow and Python, reducing manual effort by 60%.	

Additional Experience

Data Science Intern <i>Wordlane Tech</i>	Aug 2025 – Nov 2025
– Built stock market prediction models using statistical methods and machine learning, leveraging time-series analysis and mathematical feature engineering on raw financial data.	
Web Development Intern <i>Qwerty Digital Solution</i>	May 2024 – Aug 2024
– Developed and deployed client websites using WordPress and Elementor.	

Projects

DistKV – Distributed Key-Value Store <i>Go, Docker, Consistent Hashing, WAL</i>	GitHub
– Built a production-grade distributed key-value store from scratch in Go with a 4-layer architecture featuring WAL-backed persistence, consistent hashing, gossip-based failure detection, and asynchronous replication.	

- Implemented a consistent hash ring with 150 virtual nodes using SHA-256, enabling transparent request proxying and limiting key remapping to $\sim 1/N$ during node joins or failures.
- Deployed a 3-node Docker Compose cluster with heartbeat-based failure detection, automatic key rehashing and recovery, persistent WAL volumes, crash recovery via log replay, and a browser-based cluster monitor.

AI Exam Proctoring System

GitHub

CNN, OpenCV, YOLOv8

- Built a deep learning-based monitoring system for real-time cheat detection and behavior tracking.
- Enhanced detection accuracy by 27% through optimized CNN architecture and multi-person tracking.
- Automated annotation using Python & OpenCV, minimizing false positives and manual labeling.

Athena – AI Research Assistant

GitHub

LangChain, Ollama, Streamlit, FAISS

- Developed an AI-powered assistant for research paper analysis and contextual Q&A.
- Integrated RAG pipelines with FAISS & Sentence Transformers for semantic retrieval.
- Implemented multi-mode features: Summary, Q&A, Semantic Search, and Chat via Streamlit.

URL Shortener API Service

GitHub

Node.js, Express, MongoDB, Redis

- Reduced redirect latency by 45% using Redis caching.
- Achieved 3x higher throughput via Redis-based rate limiting.
- Designed a scalable backend handling concurrent requests using caching and rate limiting.

Technical Skills

- **Languages:** Python, C++, C, Go, JavaScript, Bash
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face Transformers, CNNs, YOLOv8, NumPy, Pandas
- **GenAI:** LLMs, RAG, Prompt Engineering, Fine-Tuning (LoRA, QLoRA), Embeddings, AI Evaluation, AI Guardrails
- **AI Frameworks Tools:** LangChain, LangGraph, LlamaIndex, CrewAI, FAISS, ChromaDB, Ollama, OpenAI API, Claude API, Groq API.
- **Full Stack:** React.js, Node.js, Express.js, HTML, CSS, REST APIs, FastAPI
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **Distributed Systems:** Consistent Hashing, WAL, Gossip Protocol, Asynchronous Replication, gRPC
- **Systems DevOps:** Docker, Kubernetes, Linux (RHEL, Rocky, Ubuntu), Git, GitHub Actions, CI/CD, RPM Packaging
- **Cloud Monitoring:** AWS (EC2, S3, IAM, SageMaker, Polly, Transcribe), Prometheus, Grafana, Sonatype Nexus
- **Data:** PySpark, Databricks, ETL Pipelines, Data Warehousing, Power BI
- **Core:** Data Structures Algorithms, OOP, System Design, SDLC, Agile

Certifications

IEEE INSECT 2026 Certificate | AWS Cloud Workshop | Google UI/UX Design | CheckRed Digital Defence (CDD) |